Instituto de Física da USP

Report

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Summary

1. Tasks Status
2. Learned Concepts
3. Selected Article

General Tasks Status

Task	Status	
2026-06 RETINHA presentation	◔	Link
2026-06 Scientific report	◔	Link
2026-11 Journal Article	○	Link
Dissertation	◔	Link

Task	Status	
Journal Club - Automations	◑	

Tasks Status

Task	**Status**	**Link**
Walkthrough Atlas experiment ins1759506		
↪ Centrality Bins (pt integrated)	✓	Notebook
↪ Pt Bins	✓	Notebook
↪ RAA	✓	Notebook
↪ Understand the origin of the sistematic correlated information	○	
Walkthrough Thiago's work		
↪ Scaled Spectra	◷	

Correlated Uncertainty Study Tasks Status

Task	Status	Link
Correlated uncertainty study		Notebook
$\hookrightarrow$ Model GP uncertainty prediction and compare with Atlas experiment ins1759506	◔	
$\hookrightarrow$ Check effect on Pt Scaled Spectra (Thiago's work)	◔	

Learned Concepts

Integrated Pt (Centrality bins)	✓
Pt Bins	✓
RAA	✓

Selected Article

Sequential Bayesian inference with correlated heavy-ion datasets

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Abstract

Bayesian inference provides a natural framework for updating knowledge as new information becomes available, often in a sequential manner by incorporating datasets in stages or reusing previous posteriors as priors. In practice, this is commonly implemented using a factorized update in which datasets are treated as conditionally independent. When datasets are statistically correlated, however, this approximation becomes inconsistent with the joint likelihood and can lead to biased posterior estimates. In this work, we investigate this issue in a controlled setting using pseudo-data with a tunable covariance structure. We compare joint inference, factorized sequential updating, and a formulation based on the exact conditional likelihood. We show that factorized updates reproduce the joint posterior only in the limit of conditional independence, and otherwise lead to systematic deviations that grow with the correlation strength, while conditional updates remain consistent with the joint result. To interpret these deviations, we introduce an information decomposition that separates contributions into components that are new and components that are redundant across datasets. We show that correlations induce a structured, parameter-dependent redistribution of information, governed by the overlap of dataset sensitivities. The resulting mismatch between marginal and conditional information quantitatively explains the observed deviations. These results provide a practical diagnostic for assessing the consistency of sequential Bayesian inference with correlated datasets and highlight the need for a consistent treatment of correlations within a common probabilistic framework.

Keywords: Bayesian inference, sequential updating, correlated datasets, conditional likelihood, statistical methods

Figure: Distortions of Bayesian Inference When Performed Sequentially

Results

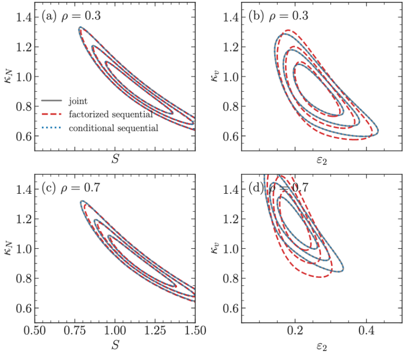

Figure 4: Posterior contours for joint inference (black solid), factorized sequential updating (red dashed), and conditional sequential updating (blue dotted), shown at $\rho = 0.3$ (top) and $\rho = 0.7$ (bottom). Left panels correspond to the (S, κ_N) plane, and right panels to the $(\varepsilon_2, \kappa_v)$ plane.

Figure: Posterior contours, for two observables, as correlation strength increases.

References I